

The while and do-while Statements

The `while` statement continually executes a block of statements while a particular condition is `true`. Its syntax can be expressed as:

```
while (expression) {  
    statement(s)  
}
```

The `while` statement evaluates *expression*, which must return a boolean value. If the expression evaluates to `true`, the `while` statement executes the *statement(s)* in the `while` block. The `while` statement continues testing the expression and executing its block until the expression evaluates to `false`. Using the `while` statement to print the values from 1 through 10 can be accomplished as in the following `WhileDemo` program:

```
class WhileDemo {  
    public static void main(String[] args){  
        int count = 1;  
        while (count < 11) {  
            System.out.println("Count is: " + count);  
            count++;  
        }  
    }  
}
```

You can implement an infinite loop using the `while` statement as follows:

```
while (true){  
    // your code goes here  
}
```

The Java programming language also provides a `do-while` statement, which can be expressed as follows:

```
do {  
    statement(s)  
} while (expression);
```

The difference between `do-while` and `while` is that `do-while` evaluates its expression at the bottom of the loop instead of the top. Therefore, the statements within the `do` block are always executed at least once, as shown in the following `DoWhileDemo` program:

```
class DoWhileDemo {  
    public static void main(String[] args){  
        int count = 1;  
        do {  
            System.out.println("Count is: " + count);  
            count++;  
        } while (count < 11);  
    }  
}
```